1. What is Computer? A computer is an electronic device that accepts data, processes it, and produces output.
2. Generations of Computers: Stages of computer development based on technology.
3. Expansions: UNIVAC – Universal Automatic Computer ENIAC – Electronic Numerical Integrator and Computer EDVAC – Electronic Discrete Variable Automatic Computer EDSAC – Electronic Delay Storage Automatic Calculator BIOS – Basic Input Output System EPROM – Erasable Programmable Read Only Memory
4. Artificial Intelligence: Ability of machines to perform tasks requiring human intelligence.
5. AI Technologies: Machine Learning, Expert Systems, Neural Networks, Robotics, NLP
6. Expert System: A system that simulates human expert decision-making.
7. Parallel Processing: Simultaneous execution of multiple tasks.
8. Microcomputer: Small computer using a microprocessor.
9. Supercomputer: Very powerful computer for complex calculations.
10. CPU Subsystems: ALU, Control Unit, Registers
11. Registers: High-speed memory inside CPU.
12. Bus: Communication pathway in a computer.
13. System Bus: Data Bus, Address Bus, Control Bus
14. Main Memory: RAM and ROM
15. RAM vs ROM: RAM is volatile, ROM is non-volatile.
16. Cache Memory: High-speed memory between CPU and RAM.
17. Input Devices: Keyboard, Mouse
18. Pointing Devices: Mouse, Trackball
19. Trackball: Stationary pointing device.
20. Joystick: Input device for games.
21. Output Devices: Monitor, Printer
22. Hard Copy vs Soft Copy: Printed output vs Screen output
23. Light Pen Uses: Selecting, Drawing, Pointing
24. Plotter: Flatbed, Drum plotter
25. Resolution & Refresh Rate: Pixels and screen refresh frequency
26. Control Unit: Controls computer operations.
27. PC vs IR: PC holds next instruction, IR holds current instruction.
28. Speech Recognition: Computer understanding speech.
29. Scanner: Flatbed, Handheld
30. Dot Pitch & Refresh Rate: Pixel spacing and refresh speed
31. Raster Scan Display: Line by line scanning display
32. Random Scan Display: Draws images directly.
33. Beam Penetration: Color generation technique.
34. Shadow Masking: Color control using metal mask.
35. Plasma Display: Gas-based flat panel display.
36. Drum Plotter: Paper wrapped around drum.
37. Flatbed Plotter: Paper fixed on flat surface.
38. Laser vs Inkjet: Toner vs Ink
39. Dot Matrix Printer: Impact printer using dots.
40. Color Depth: Bits per pixel.